

Functional Equations

Problem 1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

$$f(f(x)) = \frac{x^2 - x}{2} \cdot f(x) + 2 - x,$$

for all $x \in \mathbb{R}$. Find all possible values of $f(2)$.

Problem 2. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(y - xy) = f(x)y + (x - 1)^2 f(y)$$

holds for all real numbers x, y .

Problem 3. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 + f(y)) = (x - y)^2 f(x + y)$$

for all real numbers x and y .

Problem 4. Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f\left(\frac{x + y}{2}\right) = \frac{2f(x)f(y)}{f(x) + f(y)}$$

for all positive real numbers x and y .

Problem 5. Find all $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xy + f(x)) = xf(y) + f(x)$$

for every pair of real numbers x, y .

Problem 6. Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f\left(\frac{f(x)}{y}\right) = yf(y)f(f(x))$$

for all positive real numbers x and y .

Problem 7. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(yf(x) - x) = f(x)f(y) + 2x$$

for all $x, y \in \mathbb{R}$.

Problem 8. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the relation:

$$f(xf(y) - f(x)) = 2f(x) + xy \quad \forall x, y \in \mathbb{R}.$$